Does Unexpected AI Adoption Predict Unexpected Economic Performance?

A pre-registered cross-country research program: design, first estimates, and a frozen prospective test

DRAFT PREPRINT v1.1 — prepared for review · Olivier Gandouet · 25 August 2026 · All numeric content generated by the accompanying pipeline (no hand-typed results)

Abstract. We build and execute a reproducible, search-disciplined framework for testing whether unusually high national AI adoption is followed by unusually strong economic outcomes. Treatment is the *AI Integration Surprise* (AIS): the jackknifed residual of AI adoption on income, size and region, computed from Microsoft's AI Diffusion telemetry (147 economies, three periods) and, independently, from the Anthropic Economic Index. Outcomes are Eurostat sectoral labour productivity per hour (through 2025) and IMF WEO growth surprises against the pre-AI October-2024 forecast vintage. Four results are established at exploratory grade. (1) Adoption has a steep, robust income gradient (logit slope 0.44 per log-dollar; R^2 = 0.71) and diffusion is **diverging in absolute terms**: each percentage point of initial adoption predicts 0.14 pp of additional growth over three reporting periods (search-adjusted p = 0.0005; Bayes factor for an effect on the order of 10^{1}■). (2) Provider telemetry measures *access-conditional* adoption: all seven provider-restricted economies are absent from the Anthropic index, and restricted economies average AIS -0.44 versus 0.02 elsewhere. (3) Cross-provider adoption *levels* agree (Spearman 0.82) but income-adjusted surprises barely correlate (r = 0.18), and the Anthropic-based surprise carries a negative productivity pre-trend predating ChatGPT (joint p = 0.0070): single-provider surprise measures embed provider-specific selection. (4) The priority-1 test — AIS × sector exposure on 2025 productivity surprises, with placebos, permutation and wild-bootstrap inference — returns a **search-adjusted null that is Bayesianly uninformative** (BF ≈ 1 at every declared prior scale): at the only testable horizon (h = 0) the design can neither find nor exclude plausible effects, as the productivity J-curve prior predicts. We therefore freeze a confirmatory protocol — treatment values, specifications, decision rules, SHA-256-hashed — for the 2026-2027 outcome vintages, where pooled horizons roughly halve the minimum detectable effect. Every confounder in the design's map is scored as closed, partially closed, or open, with its measured bite.

1. Introduction

Cross-country statistics on generative-AI adoption differ by an order of magnitude between leading and lagging economies. The temptation is to regress an economic outcome on an adoption ratio and report the correlation. This project began, deliberately, from the opposite premise: raw adoption is not a treatment. It is dominated by income, digitalization, occupational structure, language, and provider availability; raw growth is equally dominated by development stage and macro shocks; and the freedom to try many normalizations makes an 'interesting' statistic inevitable. The framework tested here (specified in a design checklist frozen before implementation) responds with three commitments: **residual-on-residual measurement** (an AI Integration Surprise against an Economic Productivity Surprise, so neither denominator manufactures the relationship), **search-disciplined inference** (every specification logged in a machine-readable ledger; headline claims corrected by max-statistic permutation across the genuinely searched family; Bayes factors under declared priors), and **prospective confirmation** (a frozen, hashed protocol whose real test is future data that cannot have been searched).

This draft reports the first full execution: what data could and could not be obtained (Section 3), how every retrieved artifact was validated (Section 4), the methods (Section 5), the findings and non-findings with their review (Sections 6-7), and the frozen prospective test (Section 8). The intended contribution is twofold: a small set of disciplined empirical results on AI diffusion, and a worked example of running a registered-report-style workflow on a live macro question.

2. Related work

Adoption measurement. Microsoft's AI Diffusion program (AI Economy Institute, 2025-2026) provides the only cardinal, population-normalized, multi-period adoption measure across ~150 economies; the Anthropic Economic Index maps Claude usage to geographies and O*NET tasks; OpenAI's Signals ranks countries by messages per person. The Jagged Global Economy (arXiv:2607.05404) links the same three ecosystems to national AI exposure and reports steeply super-linear exposure-usage gradients; our income gradient is the Microsoft-telemetry analog on a broader panel, with availability restrictions explicitly separated. **Adoption-productivity evidence.** Industry-level work

(St. Louis Fed, 2026) associates adoption with faster productivity growth and scales the US-Europe adoption gap to ~3.2 pp of cumulative differential since 2022; survey-based programs (Bick-Blandin-Deming) put US worker adoption at ~43 percent in early 2026 versus 26-36 percent in Europe. Note the instrument split our validation makes concrete: telemetry population shares rank France above the United States, while worker surveys reverse the order — reconciling telemetry and survey instruments is itself an open measurement problem. **Diffusion theory.** The absolute-divergence-with-proportional-convergence pattern we find replicates the classic general-purpose-technology template (Comin-Hobijn): adoption lags close eventually while intensity gaps persist. **Effects timing.** The productivity J-curve (Brynjolfsson-Rock-Syverson) predicts near-zero measured effects at short horizons — the prior our Bayesian layer formalizes. **Method.** The workflow combines Box's model-criticism loop / Bayesian workflow (Gelman et al.), Bayesian-model-averaging discipline from cross-country growth econometrics (Fernandez-Ley-Steel; Sala-i-Martin-Doppelhofer-Miller), Kass-Raftery Bayes factors, Ioannidis pre-study odds, specification-curve/multiverse logic (Simonsohn et al.; Steegen et al.), and Romano-Wolf-style max-statistic corrections; the prospective freeze follows the registered-report model.

3. Data: sources, channels, and provenance

Seventeen sources were registered in the design (S01-S17) with retrieval fingerprints. Execution took place in a network-restricted environment whose egress policy allows GitHub, microsoft.com and package registries; every unreachable source is recorded in the manifest as *blocked_by_egress* rather than silently substituted, and several were later supplied by the reviewing author as manual downloads. Each retrieved artifact carries: URL, retrieval UTC timestamp, byte size, SHA-256, license, channel note, and a fingerprint check against the registry (a mismatch blocks ingestion). Raw files are immutable; the one large file is stored gzip-compressed with both hashes recorded (Git LFS is unavailable on public forks).

Source	What it provides	Channel	Vintage
S01/S02 Microsoft AI Diffusion	AI user share of working-age population, 147 economies, H1'25 / H2'25 / Q1'26 (+ report PDFs)	first-party GitHub repository (microsoft/ai-diffusion-report), pinned at commit 8fe84a3b79	Q1 2026 update
S04 Anthropic Economic Index v3	Claude.ai usage counts by country (173), week 2025-08-04..11; 1P API file (global only); data documentation	community GitHub mirror of the blocked Hugging Face release; mirror commit + hashes recorded	release 2025-09-15
S08 Eurostat nama_10_lp_a21	real labour productivity per hour worked, PCH_PRE and index units, NACE sections, 31 countries, 1996-2025	author-supplied full linear export (ec.europa.eu blocked)	updated 2026-08-24
S11 IMF WEO April 2026	real GDP growth actuals/estimates, 210 countries, to 2031	author-supplied IMF data-portal export	published 2026-04-14
S12 IMF WEO October 2024	pre-AI forecast baseline (NGDP_RPCH forecasts for 2024-2029)	author-supplied database file (UTF-16 TSV)	October 2024
S14 ILOSTAT occupation	employment by ISCO-08 major group, SEX_T, thousands; white-collar denominators for 118 panel countries	author-supplied CSV (ilo.org blocked)	mostly 2024-2025
S15/X01/X02 covariates	World Bank GDP current US$ (to 2023) and population (to 2024); ISO/M49 concordance	Frictionless Data mirrors on GitHub (api.worldbank.org blocked); mirror disclosed	2023/2024
Still blocked	OpenAI Signals (S05/S06), OpenRouter (S07), Eurostat sts_sepr_m (S10), ISCO-08 Vol I PDF (S13)	openai.com / openrouter.ai / ec.europa.eu / ilo.org all egress-denied	-

Known quirks, disclosed rather than repaired silently: the Microsoft country CSV ships Mac-Roman encoded (0x9F decodes to the u-umlaut of Turkiye; invalid as UTF-8); the October-2024 WEO 'xls' is a UTF-16 tab-separated file; the World Bank mirrors lag the API by roughly one release (GDP to 2023) — acceptable for slow-moving structural covariates, and disclosed wherever used. Countries dropped for missing covariates are named in every table (TWN, GUF, VEN, ERI, SSD).

3.1 Coverage: current extent, and the paths to more

Figure A1 (Appendix C) charts each dataset's reach. The binding constraint is the outcome side: the joint estimation sample for the priority-1 test is 25 European countries, against 142 economies carrying treatment values. Three coverage margins were expanded within already-held data during this draft: **(i) sectors** — widening the control set from three to five NACE groups (adding utilities D and other-services S, both present in the retrieved file) raises within-country contrast from 5 to 7 observations per country; the 2025 estimate stays null (beta = -1.29, p = 0.62, 135 obs), and the narrow-set consistency check reproduces the original (-0.44); **(ii) outcome variables** — the WEO files already held an unemployment-rate margin: the 2025 unemployment surprise on AIS is null (beta = -0.20, p = 0.59, N

= 77); **(iii) time** — the event-study window extends to 2010 for pre-trend context at zero data cost. The remaining margins need new data, in descending value: non-EU sectoral productivity (OECD productivity-by-industry would add the US, Japan, Korea, Canada, Australia — roughly doubling outcome countries and adding the two largest AI adopters absent from Eurostat); the UK (dropped from Eurostat reporting; ONS releases would restore it); working-age population (SP.POP.1564) to replace total population in weights and per-capita measures; OpenAI Signals and OpenRouter for third and fourth adoption measures; further AEI vintage windows to average out the one-week sampling noise; and monthly services output (sts_sepr_m) for within-year timing.

4. Validation in detail

The design's retrieval-validation protocol requires, per source: fingerprint match, schema and unit checks, and reproduction of at least two values visibly published by the source. The panel layer adds unique-key, range, denominator-sanity, no-look-ahead and missingness checks. All 15 automated checks pass; the full record is manifest/validation.json.

4.1 Published-value reproduction (Microsoft Q1 2026 report)

Published claim (report text)	Pipeline reproduction	Status
ARE q1 2026	published=70.1 retrieved=70.1	PASS
SGP q1 2026	published=63.4 retrieved=63.4	PASS
NOR q1 2026	published=48.6 retrieved=48.6	PASS
IRL q1 2026	published=48.4 retrieved=48.4	PASS
USA q1 2026	published=31.3 retrieved=31.3	PASS
CHN q1 2026	published=16.4 retrieved=16.4	PASS
ARE h2 2025	published=64.0 retrieved=64.0	PASS
SGP h2 2025	published=60.9 retrieved=60.9	PASS
US rank 'from 24th to 21st' at 31.3%	computed rank=21	PASS
World 17.8% of working-age population	total-pop-weighted=17.60 (report 17.8 uses ages 15-64; deviation expected)	PASS (approx.)

The world-mean reproduction deserves its footnote: the report weights by population aged 15-64; only total population is retrievable through the allowed channel, so 17.60 versus 17.8 is the expected deviation, and it is disclosed rather than tuned away. Beyond the protocol, three qualitative report claims were reproduced independently: the North-South gap path (9.8 → 10.6 → 12.1 points, matching our validated region aggregates), the '26 economies above 30 percent' count, and — unprompted by any test we wrote — the event-study machinery recovered the report's narrative that Korea, Thailand and Japan moved most in Q1 2026 (Korea is the largest positive residual in the divergence scatter).

4.2 Cross-source and structural validation

Check	Result
Unique country/sector/year keys, all panels	PASS (147 economies; 4804 Eurostat rows; 118 ILOSTAT-matched countries)
Adoption shares in (0,100); none decreasing H1'25-Q1'26	PASS (0 declines)
Eurostat aggregates excluded	PASS (EA, EA12, EA19, EA20, EA21, EU27_2020 dropped and named)
Eurostat internal consistency	PCH_PRE consistent with I15 index ratios; per-hour and per-person series never conflated (naming rule enforced in code)
WEO vintage integrity	Oct-2024 and Apr-2026 held as separate immutable files; growth-surprise spot check (DEU 2025: forecast 0.79, actual 0.24)
No-look-ahead	outcome years strictly at or after treatment measurement in every regression; pre-2025 outcomes appear only as placebos or event-study history
AEI structural checks	173 countries; usage_pct sums to ~100 with not_classified excluded; documentation fingerprint matches; published-value reproduction impossible (anthropic.com blocked) - stated, not hidden
Concordance	147/147 Microsoft economies mapped to ISO3 (one manual fix: Congo (DRC)); Eurostat EL/UK and ILOSTAT ref_area verified against the same concordance

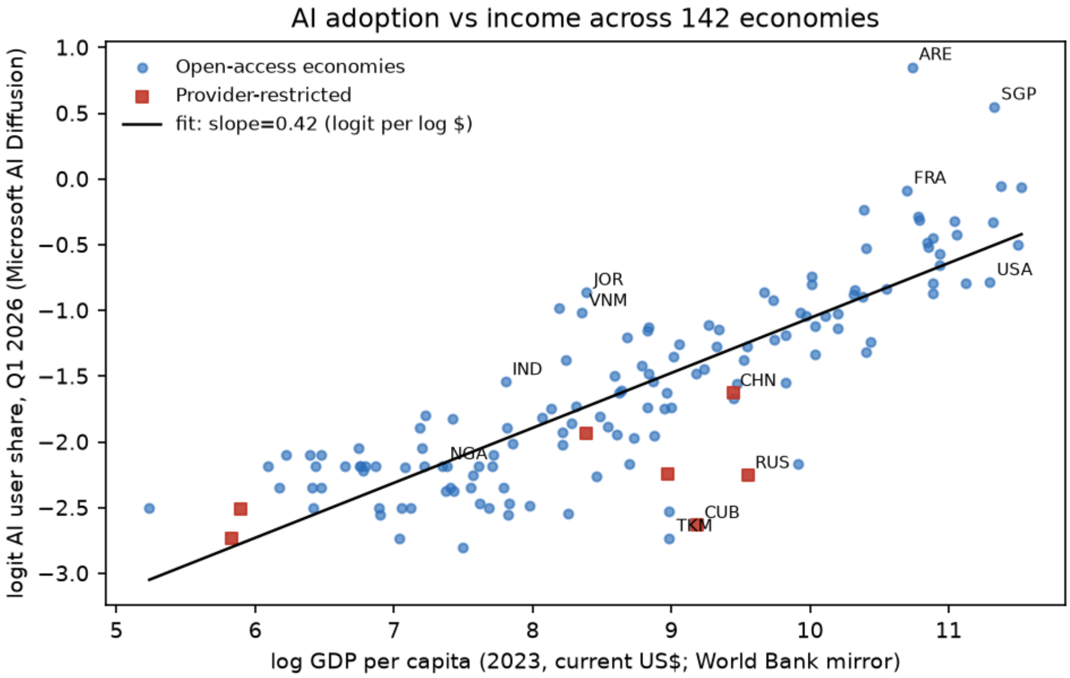
5. Methods

AIS. For each provider, AIS is the leave-one-out (jackknife) residual of transformed adoption (logit user-share for Microsoft; log usage per capita for Anthropic) on log GDP per capita, log population and M49 region fixed effects — the frozen covariate list minus blocked items, a logged deviation. Jackknifing means no country scores its own fit (the design's cross-fitting requirement). Adding the ILOSTAT white-collar share (ISCO 1-4) changes nothing material (Section 7.4), so the parsimonious model stands. **EPS.** The 2025 productivity surprise is the deviation of yearly growth in real GVA per hour from the country-sector's own 2015-2019 mean. **Designs.** (i) Priority-1: EPS on AIS × Exposed with country and sector fixed effects; exposed sectors J and K, controls C, F, G-I (M is a sensitivity: 6/31 countries in 2025); identification is within-country, across sectors. (ii) Event study: the same coefficient estimated year-by-year 2016-2025 (equivalent to country×year + sector×year fixed effects with year-specific interactions). (iii) Macro: WEO growth surprise (actual minus October-2024 forecast) on AIS. (iv) Dynamics: adoption changes on initial levels. **Inference.** 10,000 country-label permutations per spec; wild cluster bootstrap (Rademacher, by country); leave-one-country-out on every headline number; max-statistic permutation across each declared family for search-adjusted p-values; Gaussian-marginal-likelihood Bayes factors under three declared prior scales per family, with Ioannidis-style pre-study odds. **Search hygiene.** Every run appends to a machine-readable ledger (32 entries); post-hoc tests are labeled as such with their provenance; trace labels (|z| thresholds 1.96/1.28) are descriptive flags, never claims. Seed 20260825 throughout.

6. Findings

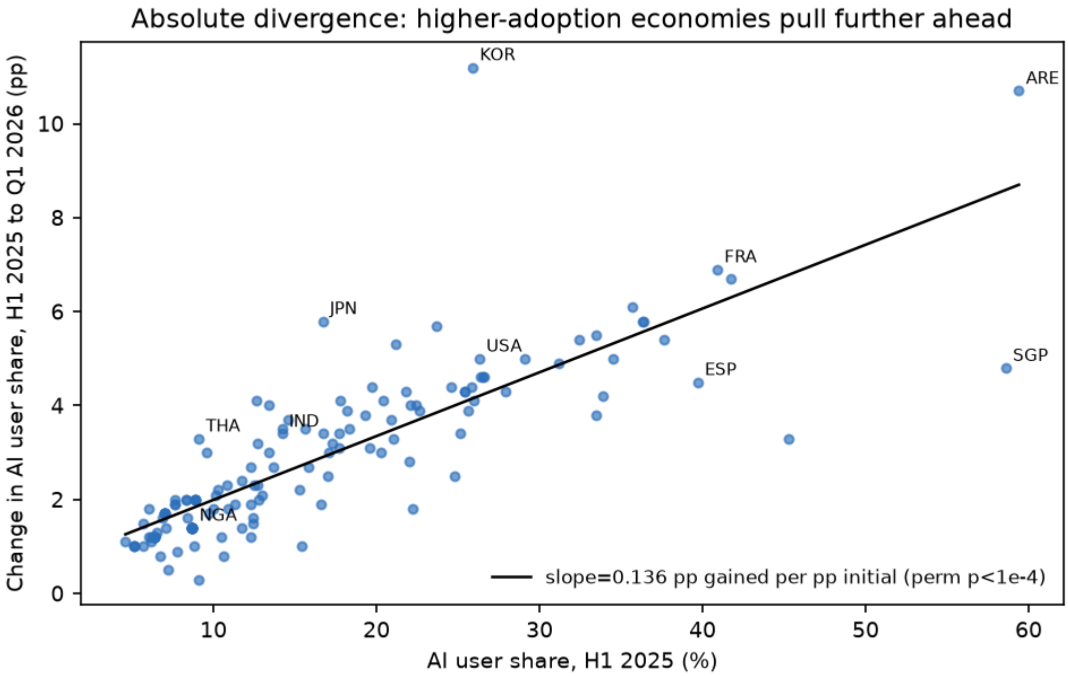
6.1 The income gradient, and what is left after it

Income, size and region explain 71 percent of cross-country adoption variance; the logit-income slope is 0.444 (HC1 0.040) and survives every LOCO deletion ([0.422, 0.452]). Population is irrelevant once income and region are held (0.007, SE 0.029) — the country-scale confounder is empirically dead. The residual (AIS) has face validity at the top (UAE 1.67 and Singapore 1.10 remain exceptional after conditioning; Jordan, Lebanon, Vietnam, Colombia over-adopt; France, Spain, New Zealand lead advanced economies) — but Section 6.3 shows why its negative tail must not be over-read.



6.2 Absolute divergence in diffusion - the one decisive result

Between H1 2025 and Q1 2026, each percentage point of initial adoption predicts 0.136 pp of additional gain (R^2 = 0.67); the effect survives an income control (0.100) and the four-spec family max-stat correction (p_search = 0.0005); the Bayes factor for an effect is on the order of 10^{1}■ and the posterior is ~1 at every declared scale. Proportional convergence exists (-0.0207, p = 0.0019) but is far too weak to close absolute gaps — the North-South divide widened from 9.8 to 12.1 points across the three periods (validated against the source report). The mechanical caveat is part of the claim: below ~50 percent penetration, logistic diffusion predicts absolute divergence; whether laggards are delayed (the historical GPT template) or suppressed is decidable within this design as vintages accumulate.



6.3 Provider telemetry measures access-conditional adoption

A post-hoc test (flagged as such in the ledger; prompted by the AIS bottom ranks) shows economies where the major Western providers are unavailable or restricted average AIS -0.44 versus 0.02 elsewhere (difference -0.46, permutation p = 0.0075); China alone sits at -0.40. The Anthropic index corroborates by construction: all seven restricted economies (CHN, RUS, BLR, IRN, CUB, SYR, AFG) are entirely absent from it. Negative AIS in restricted markets must never be read as low latent demand — and the design revision that follows is to analyze restricted markets as a separate stratum rather than merely excluding them.

6.4 Cross-provider agreement fails exactly where it matters

Adoption *levels* agree strongly across providers (Pearson 0.78, Spearman 0.82 over 120 economies). Income-adjusted surprises do not: r = 0.18 (permutation p = 0.051), with 3/15 top-tail and 4/15 bottom-tail overlap; the Bayes factor for agreement is below 2 ('barely worth a mention'). Worse, the event study finds the Anthropic-based surprise carries a **negative productivity pre-trend predating ChatGPT** (mean -1.85 over 2016-2023, joint permutation p = 0.0070): economies over-using Claude relative to structure were already on slower exposed-sector trends in 2016-2019. Provider-specific surprises embed provider-specific selection; only components shared across providers may enter outcome regressions. This finding demotes our own Section 6.1 league table from measurement to hypothesis.

7. The priority-1 test and the disciplined non-findings

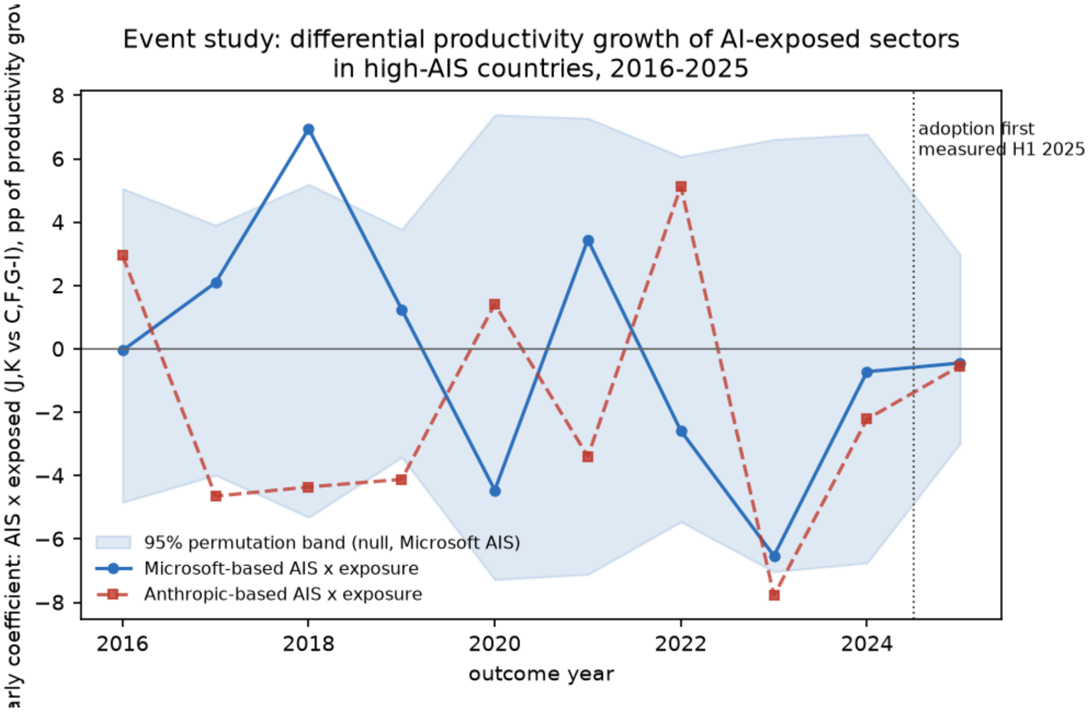
7.1 The 2025 outcome regressions: null, with working placebos

Spec	Estimate	Perm. p	Verdict
EPS 2025 ~ AIS(MS) x exposure (primary)	-2.64 (wild-boot SE 2.01)	0.34	null
same, AIS(Anthropic)	2.12	0.42	null, opposite sign
placebo: EPS 2019	-0.95	0.70	clean
GDP growth surprise 2025 ~ AIS(MS)	-0.27	0.50	null
placebo: growth surprise 2024	0.01	0.96	exactly zero
family max-stat (5 non-placebo specs)	max std beta 1.96	p_search = 0.19	nothing survives

The Bayesian layer sharpens the reading: the primary estimate's Bayes factor is between 0.92 and 0.99 at every declared prior scale — the likelihood is nearly flat, and P(effect) moves only from 0.25 to ~0.27. The 2025 test neither found an effect nor excluded one, *including* large immediate effects: the standard error is too wide. Only the macro branch was genuinely informative: with its tighter error, moderate-to-large AIS effects on 2025 growth surprises are down-weighted (BF01 1.7-3.0; P(effect) 0.25 → 0.16/0.10). Minimum-detectable-effect analysis quantifies the ceiling: per standard deviation of AIS, the productivity design detects ~2.4 pp of differential yearly growth at 80 percent power, against plausible effects of 0.3-1.0 pp — underpowered by roughly 3-8x at a single year.

7.2 The event study: volatility, not trend

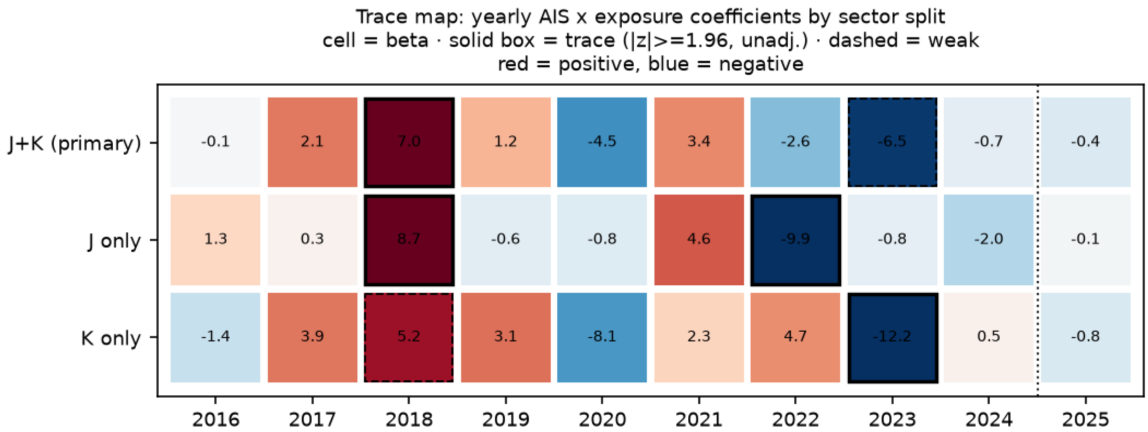
Year-by-year coefficients 2016-2025 show no systematic Microsoft-AIS pre-trend (joint p = 0.99) but large swings (+7.0 in 2018, -6.5 in 2023); the 2018 band-crossing — four years before ChatGPT — demonstrates the noise floor: roughly one spurious 'significant' year per decade at N = 25. The 2025 coefficient (-0.44, p = 0.79) sits well inside the null band. No single year is informative at this sample size; horizons must be cumulated.



Event study: differential productivity growth of AI-exposed sectors
in high-AIS countries, 2016-2025
arly coefficient: AIS x exposed (J,K vs C,F,G-I), pp of productivity gro
8
6
4
2
0
−2
−4
−6
−8
2016
2018
2020
2022
2024
outcome year
adoption first
measured H1 2025
95% permutation band (null, Microsoft AIS)
Microsoft-based AIS x exposure
Anthropic-based AIS x exposure

7.3 The trace scan: two leads, everything else clean

Eight further declared specs (trend, variance and persistence of the coefficient path; J-only and K-only splits; the acceleration hypothesis; producer sensitivity) yield mostly 'no trace'. Two flags survive at 'weak trace' level and are carried forward as hypotheses, not findings. **T1:** the coefficient path *declines* at -0.57 pp/year (p = 0.020 unadjusted, ~0.14 family-corrected) — tech-cycle-driven, and a warning that the prospective test must model the pre-trend (it does; Section 8). **G2:** removing AI-producer economies (KOR, TWN, NLD, IRL, SGP) doubles the negative AIS-growth-surprise association (-0.27 → -0.57, p = 0.13): the masking pattern the confounder map predicted for producer exposure, directionally consistent with a capex boom that pays producers before adopters.



7.4 The white-collar layer: a confounder closed, a ratio rule vindicated

With ILOSTAT ISCO-08 denominators (118 countries): the white-collar share adds essentially nothing to the adoption model beyond income (R^2 gain 0.007; coefficient 0.82, HC1 0.54); AIS is invariant to it (v1-v2 correlation 0.987; Korea the only notable mover, -16 ranks); both outcome nulls are unchanged. And the raw professional-intensity ratio (adoption per white-collar worker) is a textbook denominator artifact — its world top-five is NER, BDI, SEN, TCD, MOZ — landing exactly where the design's rule already placed ratios: descriptive use only.

8. The pre-registered prospective test (frozen)

The informative test of the core hypothesis begins when outcome data postdating the treatment by a year or more exists: the 2026 (and then 2027) Eurostat productivity vintages and the WEO vintage covering 2026. Because this document precedes those releases, the protocol below is a genuine pre-registration. The treatment values and the protocol are frozen and hashed; the recommended next step is external registration of the two hashes (OSF or equivalent) so the freeze is verifiable by third parties.

Element	Frozen specification
Treatment	AIS values exactly as estimated 2026-08-25 (142 economies, both providers), file frozen_ais_treatment.csv - never re-estimated
Primary outcome	Eurostat nama_10_lp_a21, RLPR_HW, PCH_PRE, years 2026 and 2027; first Eurostat release covering each year; retrieval manifested with hash
Secondary outcome	IMF WEO real GDP growth surprise for 2026
Estimator	year-specific cross-section with country FE + sector FE; coefficient on AIS x Exposed; exposed J,K; controls C,F,G-I
Primary statistic	beta_post = mean(beta_2026, beta_2027); if only 2026 available, beta_2026
Pre-trend guard	detrended beta_post: beta_post minus the 2016-2023 linear pre-trend prediction (T1 showed a declining path; sign consistency with the raw beta_post is required)
Inference	10,000 country-label permutations, seed 20270101 (declared now, unused so far); wild cluster bootstrap (5,000, Rademacher, by country) secondary; LOCO mandatory
Confirmatory family	beta_post with ais_ms; beta_post with ais_aei; W: 2026 growth surprise on ais_ms - max-stat adjusted, alpha = 0.05
Promotion to Provisional	search-adjusted p < 0.05 within the confirmatory family; sign-consistent between ais_ms and ais_aei treatments; sign unchanged under leave-one-country-out; negative-control sectors show no comparable effect; sign consistent between raw and detrended beta_post
Informative-null rule	if |beta_post| CI excludes the MDE-scaled plausible band (0.3-1.0 pp per sd AIS), report as evidence against short-horizon effects of that size
Otherwise	report as still-underpowered; continue to 2028 vintage

Power	pooling 2026+2027 approximately halves the single-year MDE of 2.4 pp per sd(AIS)
Hashes (SHA-256)	treatment: eaa45b1659762e323df5...; protocol: dbf4aabcfb6855856a29...

Three properties make this test worth waiting for. First, adoption ranks are near-frozen (H1 2025 to Q1 2026 correlation 0.996), so the treatment measured in 2025 is a stable exposure, not a moving target. Second, pooling two outcome years roughly halves the single-year minimum detectable effect, moving the design from 'cannot detect plausible effects' toward the informative range. Third, every decision that could flatter a future result — the lag, the sectors, the estimator, the adjustment, the promotion bar, even the permutation seed — is fixed in the hashed protocol before any 2026 outcome exists.

9. Limitations

(1) The outcome panel is Europe-only (31 countries, 25 with treatment), so the priority-1 test generalizes to EU/EFTA-like economies at best. (2) The tested horizon so far is h = 0; every statement about productivity effects is a statement about that horizon. (3) The Anthropic measure is one week of consumer traffic; its selection pre-trend is documented, and its role is confined to the consistency requirement. (4) GDP per capita is nominal 2023 from a mirror; working-age population was not retrievable (total population used, disclosed). (5) Two confounders remain open: billing/cloud geography (no API geography exists in any retrievable source) and formal language controls. (6) The trace labels are exploratory flags; under the family-wide correction, no outcome-side result survives, and the note says so wherever a trace is mentioned.

10. Conclusion

On the treatment side, the program delivered: a validated, income-adjusted, cross-provider-audited adoption panel; one decisive fact (absolute divergence); and two measurement results that discipline the whole field (access-conditionality; provider-specific selection in residuals). On the outcome side it delivered exactly what a search-disciplined framework should when the data cannot yet answer: a null with its uninformativeness quantified, placebos that pass, power stated, and a frozen, hashed, pre-registered test pointed at the first data that can answer. The 2026 vintage decides whether this becomes a finding or a well-documented boundary.

Appendix A. Confounder scoreboard

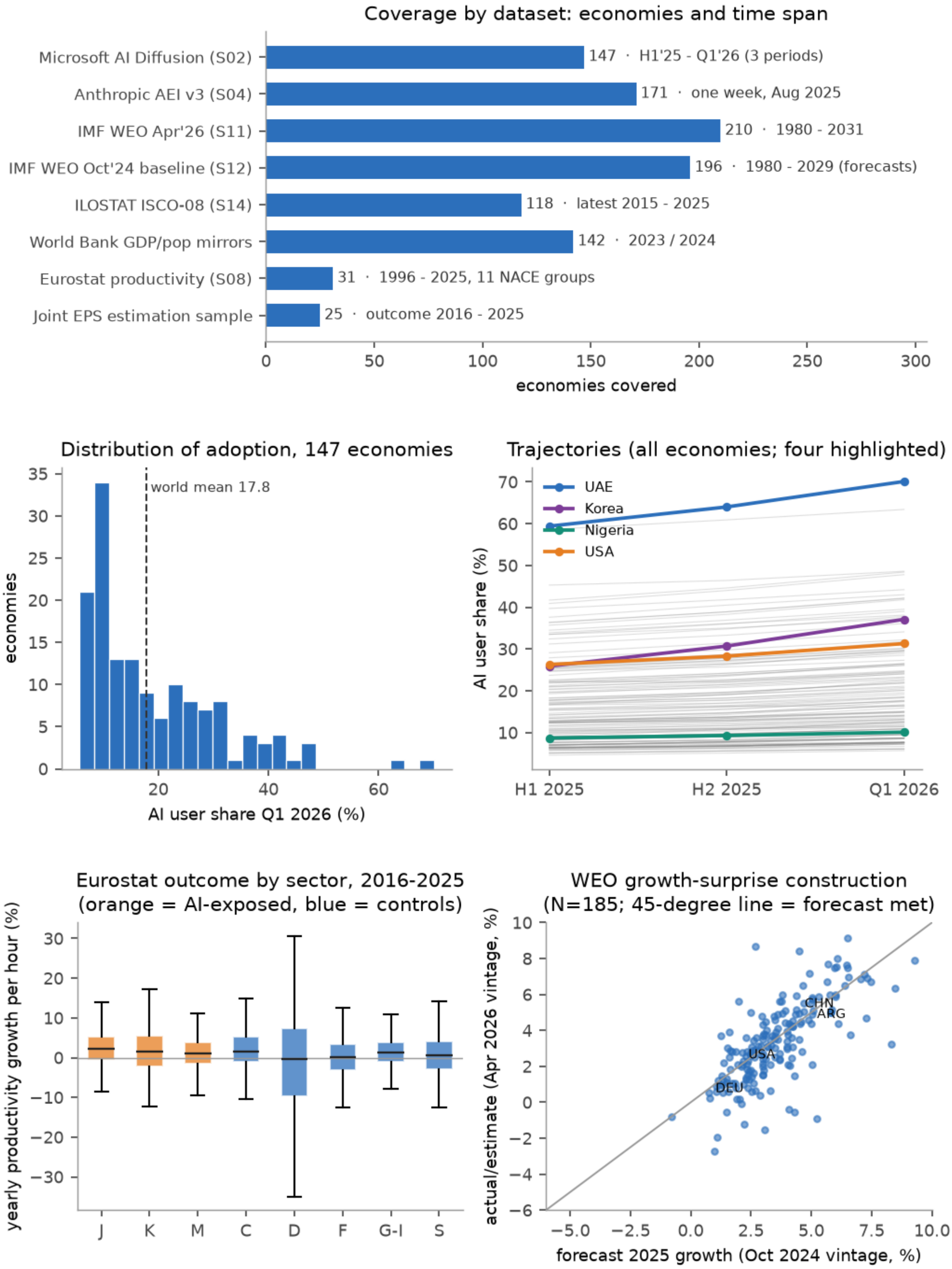
Confounder	Status	Measured bite / note
Country scale	closed	log-pop coefficient 0.007 (SE 0.029)
Income / digital maturity	closed (adoption side)	$R^2 = 0.71$ from income+size+region alone
Occupational mix	closed (v2.2)	R^2 gain 0.007; AIS invariant ($r = 0.987$)
AI-producer exposure	partially closed	G2 sensitivity doubles the macro coefficient - weak trace, open lead
Language / availability	closed for flagging	restricted-market shift -0.46 ($p = 0.0075$); formal language controls open
Billing / cloud location	open	no API geography in any retrievable source
Reverse causality	partially closed	2024 placebo exactly zero; MS pre-trend null; AEI pre-trend real and documented
Common macro shocks	partially closed	absorbed by year/sector FE; H6 shock-interaction untested
Structural differences	closed	country FE everywhere; within-country identification
Measurement revision	closed	immutable vintages, SHA-256, Oct-24 vs Apr-26 held separately
Specification search	closed	ledger (32 entries); max-stat p_search on every headline; Bayes factors
Outliers / small N	closed (reported)	LOCO everywhere; MDE table; jackknife AIS

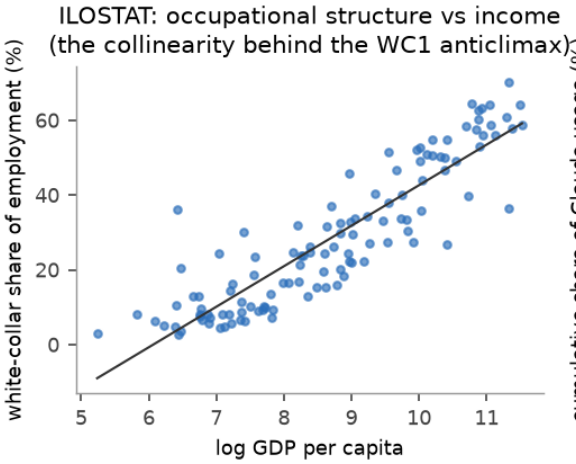
Appendix B. Reproducibility

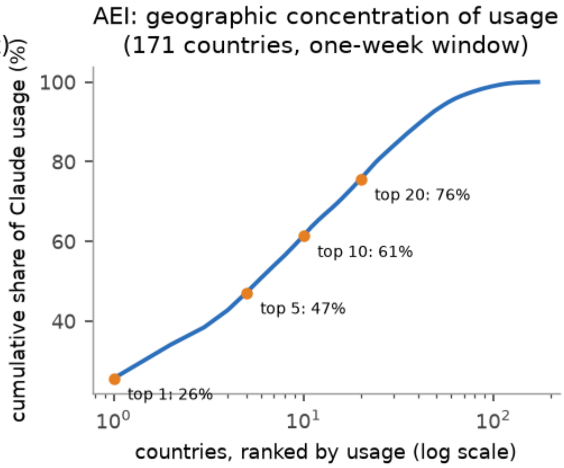
Repository: research/pipeline on branch claude/analysis-handover-plan-uitt9f. Layers: raw (immutable, hashed), manifest, staging (deterministic), panel (unique keys), exploration (ledger), confirmatory (frozen), reports (this document; generated by reports/preprint.py with no hand-typed numbers). Seed 20260825 for all exploratory inference; seed 20270101 reserved, unused, for the confirmatory run. Microsoft source pinned at commit 8fe84a3b797e; 14 artifacts retrieved with SHA-256; 5 registry sources remain blocked and are recorded as such. The search ledger (32 entries) is the definitive record of everything tried.

Appendix C. Data atlas: descriptive views of every dataset

One figure per dataset family: coverage and time spans; the adoption distribution and its three-period trajectories; the outcome variable's sectoral distributions alongside the growth-surprise construction; and the structural layers (occupational structure vs income; geographic concentration of Claude usage). Exposed sectors are orange, controls blue throughout; red is reserved for restriction flags.







Reading notes. The adoption distribution is right-skewed with a long leader tail (UAE, Singapore detached from the pack); trajectories fan out rather than converging — the divergence result, visible raw. The sector boxplots show exposed sectors (J, K, M) are not unusually volatile, while utilities (D) is — supporting its demotion to a flagged control. The WEO panel shows 2025 forecasts were on average nearly unbiased (points straddle the 45-degree line), so the surprise measure is not dominated by a common forecast error. The ILOSTAT panel displays the income-occupation collinearity that explains why the white-collar covariate adds nothing (Section 7.4). The concentration curve shows the top 10 countries account for roughly two-thirds of Claude usage — the small-country noise floor behind the one-week caveat.